

Temperature Monitor Project

Arduino Uno + TMP Sensor (Power / Vout / GND) + LED Alert

Components Used

Component	Purpose
Arduino Uno R3	Brain of the circuit; reads sensor data and controls the LED
TMP Temperature Sensor	Converts ambient temperature into a voltage (analog output)
LED (amber)	Lights up as a visual alert when temperature crosses the threshold
Resistor (220 ohm)	Limits current to the LED so it does not burn out
Jumper wires	Red = 5V, Black = GND, Orange = A0 (Vout), Green = Pin 13 (LED)

TMP Sensor Pinout (Tinkercad labels)

Pin 1	Power	-> Arduino 5V
Pin 2	Vout	-> Arduino A0
Pin 3	GND	-> Arduino GND

Build Steps

- 1. Open Tinkercad** — Start a new circuit in Tinkercad Circuits.
- 2. Add components** — Drag in Arduino Uno, TMP Sensor, LED, and a Resistor.
- 3. Wire the TMP sensor** — Power -> 5V, GND -> GND, Vout -> A0 (as shown in the diagram).
- 4. Wire the LED** — Anode -> Pin 13, Cathode -> Resistor (220 ohm) -> GND.
- 5. Add the code** — Paste the sketch below into the Tinkercad code editor.
- 6. Run the simulation** — Click 'Start Simulation', click the TMP sensor to change the temperature, and watch the LED and Serial Monitor.

Full Arduino Code

```
// Temperature Monitor Project
// TMP Sensor (Power / Vout / GND) + LED Alert

int tempPin = A0;      // TMP sensor Vout pin
int ledPin = 13;       // LED alert pin

float temperatureC = 0;
float voltage = 0;

float threshold = 30.0; // agar temp isse zyada ho to LED jal jaye (Celsius)

void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {
  int sensorValue = analogRead(tempPin);

  // Analog value ko voltage me convert karo
  voltage = sensorValue * (5.0 / 1023.0);

  // Voltage ko Celsius me convert karo
  temperatureC = (voltage - 0.5) * 100;

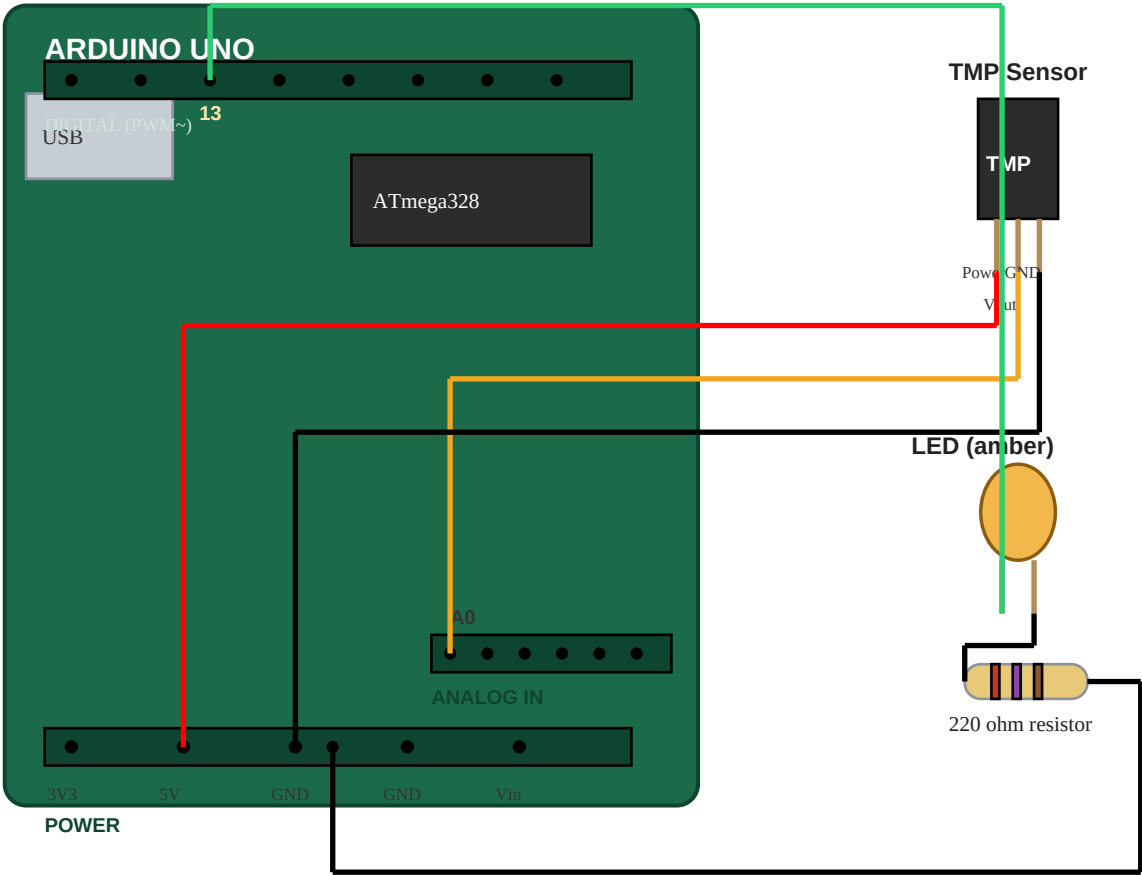
  Serial.print("Temperature: ");
  Serial.print(temperatureC);
  Serial.println(" C");

  if (temperatureC > threshold) {
    digitalWrite(ledPin, HIGH); // Alert ON
    Serial.println("ALERT! Temperature too high");
  } else {
    digitalWrite(ledPin, LOW);   // Alert OFF
  }

  delay(1000);
}
```

Circuit Diagram

White background, color-coded wiring for easy reference while building in Tinkercad.



Wiring Summary

Wire Color	From	To
Red	TMP Power	Arduino 5V
Black	TMP GND	Arduino GND
Orange	TMP Vout	Arduino A0
Green	Arduino Pin 13	LED Anode
Black	LED Cathode -> Resistor	Arduino GND

Tinkercad Circuit Screenshot

